HCA-4000: High Conductivity Adhesive

Deep Materials

Product Description

HCA-4000 is a thermally conductive adhesive tape designed to provide efficient heat transfer and reliable bonding between heat-generating components and heat sinks. The material incorporates thermally conductive, electrically insulating fillers to deliver low thermal resistance while maintaining excellent electrical isolation.

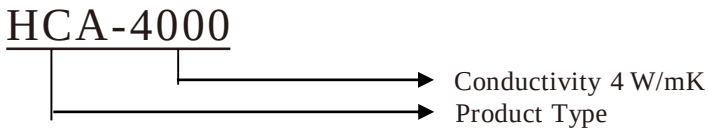
Its low modulus formulation enables effective gap filling and surface conformability, minimizing mechanical stress on sensitive components during assembly and operation.

HCA-4000 is fully cured, chemically inert, and stable under harsh environmental conditions, providing long-term reliability without contamination. The material is silicone-free with low oil bleed, ensuring compatibility with sensitive electronic applications.

Applications

- Bonding of heat sinks to power devices, processors, and other heat-generating components
- Consumer and computing electronics, including SSDs
- Power electronics modules requiring electrical isolation
- Automotive and industrial
- Systems requiring adhesive-based thermal interface solutions with low mechanical stress

Product Identification



Application Guidelines

- Avoid exposure to open flame
- Ensure application surfaces are clean

Key Features

- Low thermal resistance for efficient heat transfer
- Self-adhesive for simplified assembly
- Low modulus formulation for reduced mechanical stress
- Excellent electrical insulation
- Silicone-free with low oil bleed for contamination control

Storage and Shelf Life

- Store at 10°C to 30°C in a dry environment.
- Shelf life is 12 months from the date of manufacture.

Packaging Information

- Available in sheet form or die-cut to customer-specified dimensions and thicknesses

Property	Value	Test Method
Color	White	Visual
Thermal Conductivity (W/m·K)	4.0	ASTM D5470
Minimum BLT (μm)	40	--
Density (g/cc)	3.25+/-0.1	ASTM D792
Dielectric Strength (kV/mm)	>8	ASTM D149
Operating Temperature Range (°C)	-50 to 120	--
Volatility (%)	＜0.01	ASTM E595
Flammability Rating	V-0	UL94
Dielectric Constant (1 GHz)	7.5	ASTM D150
Viscosity (Pa·S)	900+/-200 @ 60°C	
Shear strength (Mpa)	0.3 @ 25°C	ASTM D1002 Al/Al

The data provided are typical values derived from laboratory testing and are intended for reference only. They should not be used as specifications or as a basis for design without independent validation.